Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building AI-powered workflows, intuitive product experiences, data pipelines, and scalable cloud systems. Experienced in Python, TypeScript, React, REST APIs, SQL, document processing, analytics, automated testing, and rapid product delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and analytics dashboards for 500+ users, translating complex data into intuitive product workflows while reducing response latency from 1.2 seconds to 350 milliseconds.
- Implemented fault-tolerant pipelines for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving observability, data integrity, and reliable production operations.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed responsive React interfaces and Flask REST services for document upload, routing, status, and history workflows, translating user feedback into maintainable features supporting 150+ users and 10,000+ daily requests.
- Created concurrent ingestion for 1,000+ files with automated validation, AWS RDS persistence, and structured error handling, reducing processing turnaround 60% while maintaining reliable data workflows during peak demand.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible machine-learning and data pipelines processing 100,000+ records, applying automated evaluation, feature instrumentation, and systematic debugging to improve long-horizon forecasting accuracy 18% across controlled experiments.
- Collaborated on model design and validation experiments, documenting technical decisions and tuning a spatial-temporal architecture that improved closed-loop prediction stability 15% across repeated controlled forecasting simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed REST services and multi-source ETL pipelines using Python, SQL, AWS Glue, and S3, improving response times 30% and data-processing efficiency 68% across complex customer financial workflows.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across three platforms, accelerating debugging, feedback cycles, and overall engineering confidence in rapid production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js product experiences across eight customer, billing, permissions, and notification workflows, synchronizing 2 million+ Azure SQL records through maintainable services and real-time WebSockets.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving iteration speed, feature quality, and overall operational reliability.

Skills

Languages & Frontend: Python, TypeScript, JavaScript, Java, SQL, React, Next.js, HTML5, CSS3, Tailwind CSS, responsive interfaces, data visualization
Backend & AI: REST APIs, Flask, FastAPI, Node.js, microservices, ETL pipelines, document processing, LLaMA-2, FAISS, vector search, NLP, forecasting, workflow automation
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, AWS RDS, AWS Glue, Amazon S3, Azure Blob Storage, data modeling, indexing, query optimization
Quality & Delivery: Git, Docker, Kubernetes, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, debugging, observability, Agile, product collaboration, rapid iteration

Projects

FinNexus | Live | *TypeScript, React, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built intuitive financial workflows supporting 50,000+ users and one million transactions, integrating secure APIs, real-time synchronization, advanced analytics filters, MongoDB persistence, Kubernetes, and automated CI/CD deployment.

Big-Brain | Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, REST APIs, GCP*
- Built an AI document-processing platform using LLaMA-2 and FAISS, delivering semantic extraction, search, and summarization for 50,000+ documents with 95% query accuracy and real-time collaborative workflows.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, Docker, Maven*
- Built an AI-assisted data workspace supporting 10+ databases, combining Java services, TypeScript interfaces, schema-aware query generation, analytics dashboards, Docker, and Maven for reliable financial exploration and management.